

NGR Operating Architecture Proposal v2 Protected Manual and Adaptive Cruise Control (ACC)

This revision replaces “Protected Automatic” with Adaptive Cruise Control (ACC) and incorporates the May 16, 2026 refinements.

Operating Modes

Manual – human authority, no railroad protection.

Protected Manual – human authority, railroad protection, reverse allowed, human restart after holds.

ACC – light automation with operator override, forward only, automatic restart after holds.

Auto – CTO / CE choreography.

Railroad Truth Model

Peer table = occupancy authority. A locomotive occupying a block operationally protects that block. Separate dispatcher work-zone closure is removed for v1.

ACC Principles

- Forward only; reverse requests denied.
- Operator selects speed; CTO cruise used only if none selected.
- ACC speed capped conservatively (initial target 100–110 PWM).
- No routine station stopping.
- Continues indefinitely until protection intervenes.
- First magnet pair arms protection and joins peer tables.

Protection Behavior Shared by Protected Manual and ACC

- Magnet acquisition
- Peer broadcasting
- Occupancy enforcement
- Restricted approach behavior
- Stop enforcement at block boundary

Restricted Approach

If next block is occupied or uncertain, train may continue at restricted speed (~PWM 60). If still unsafe at next boundary, stop using closer approach profile (~50).

Operator Interface

ACC LED = active.

Messages only explain denied actions: ACC requires forward, reverse blocked, block occupied, position unknown, etc.

ACC Philosophy

ACC is adaptive road running: operator requests speed, railroad modifies movement, automatic resume allowed, operator may reclaim authority.